Zixian Liu

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Education

Purdue University West Lafayette, IN
B.S. in Computer Science; Minor in Mathematics (GPA: 3.94/4.0) *Jul. 2023 – May 2027 (expected)*

- **Selected coursework:** Machine Learning, Algorithms, Operating Systems, Numerical Methods, Linear Algebra, Statistics.

Manuscripts Under Review

[1] First-author manuscript on real-world robot learning and manipulation, under review at **IEEE ICRA 2027**. *(Title omitted during double-anonymous review.)*

Research Experience

Machine Learning and Robotics Research, Prof. Yung-Hsiang Lu Jan. 2026 – present
Purdue University

Multimodal Robot Learning with Action-Space Expansion and Cross-String Adaptation Aug. 2026 – present

- Developing a policy architecture that integrates multimodal sensory observations, scores from a supervised human-preference model, and musical score requirements to generate bowing commands for a UR5e-based robotic cellist.
- Investigating multimodal reward design and evaluation using audio, force/torque, vision, and robot state, with independent listening studies to assess performance.
- Extending the bowing action space from speed and depth to include bow angle and contact position.
- Conducting experiments to assess how learned bowing policies generalize across cello strings and adapt to string-specific contact conditions.

Residual RL for Precise, Contact-Rich Manipulation [1] Jan. 2026 – Sep. 2026

- Developed a four-stage residual SAC framework for a UR5e robotic cellist under contact drift, using a fixed reward classifier trained on human ratings.
- Designed held-out classifier evaluation and a blinded listening study. Policies achieved a 15 % mean held-out classifier-score gain over the planner across five pieces.
- Built the real-robot training and safety stack; wrote the first-author manuscript and an audit reproducing reported results from recordings and logs.

Team lead since Summer 2026; coordinate five undergraduates across two research tracks in experiment planning and results analysis; attend the weekly graduate research meeting.

Vision-Based Hand-Gesture Conducting

- Built a computer vision interface with MediaPipe's pretrained hand model and OpenCV for camera-based hand tracking and beat extraction.
- Prototyped real-time tempo estimation and experimental phase-following audio feedback, with trajectory visualization, tracking-loss recovery, and synthetic timing tests.

Selected Projects

Xinu OS Aug. 2026 – present

- Implemented semaphore-based synchronization and an independent IPC extension with bounded message buffers and blocking send/receive. Deployed and tested Xinu on ARM-based BeagleBone Black hardware.
- Implemented a kernel-level semaphore checker to detect inconsistencies in semaphore counts, waiting queues, and process states.

Unix Shell Jan. 2026 – May 2026

- Built a bash-style shell in C/C++ with a lex/yacc parser, supporting pipes, redirection, background jobs, subshells, signal handling, variable and wildcard expansion, and command history.

Web Crawler and Search Engine Aug. 2025 – May 2026

- Developed a web crawler and search engine with indexing, hash-table and AVL-tree data structures, and multi-keyword queries.
- Benchmarked query performance across data-structure implementations and continued maintaining the project beyond the course.

Systematic Trend-Timing Strategy Feb. 2026 – Sep. 2026

- Built a vectorized backtester using 100 years of daily data, modeling next-open execution, transaction costs, and taxes.
- Evaluated 233 strategy variants using pre-registered walk-forward validation, block bootstrap, and multiple-testing correction; identified selection overfitting and reduced crisis drawdowns without excess returns.